Constructive local–global passage by finite trees

Certificates, the first-unit scan, and standard étale charts

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Part I

The mathematical argument

1 The output determines the useful local statement

Let $A \to E$ be a map of commutative rings. The desired global description consists of finitely many $s_i \in E$, coefficients $a_i \in E$, and standard étale presentations of $E[1/s_i]$, with

$$\sum_i s_i a_i = 1.$$

Thus an appropriate local output is a finite algebraic certificate for one such chart, together with information allowing its denominator to participate in a finite cover. Merely asserting the existence of a chart at each existing prime is a different kind of input.

The following construction works intuitionistically. It uses neither prime-ideal existence nor excluded middle. Its input is an inductively constructed finite derivation, not an arbitrary pointwise assertion about primes. Producing that derivation from a particular local proof is part of the work.

2 Radical/unit branches

Fix a ring R. A state consists of two finite lists I and U of elements of R. Write (I) for the ideal generated by the first list, $M(U)$ for the monoid generated by the second, and put

$$S(I;U) = M(U) + (I).$$

Here membership in the sum retains witnesses. This multiplicatively closed set contains 1. In $R_{S(I;U)}$ the elements of U are units and the extension of (I) lies in the Jacobson radical.

A test of an element $a \in R$ produces two successors:

$$(I, a; U) \qquad \text{and} \qquad (I; U, a).$$

The first successor puts a in the radical ideal; the second makes it a unit. Both successors are retained. This construction does not assert that one can decide whether a is a unit in the parent ring.

Lemma 1 (One branching step)**.** *If $x \in S(I, a; U)$ and $y \in S(I; U, a)$, there is $t \in S(I; U)$ lying in the ideal (x, y).*

Proof. Write $x = s + ab$, where $s \in S(I;U)$, and $y = va^k + j$, where $v \in M(U)$ and $j \in (I)$. Then

$$t = s^k v + (-b)^k j$$

belongs to $S(I;U)$. Modulo (x), $s = -ab$, so $t = (-b)^k y$ modulo (x). Expanding the difference of powers gives an explicit expression for t in (x, y). □

Theorem 2 (Certified finite cover)**.** *Start at the empty state and form a finite binary tree of radical/unit branches. At every leaf $(I;U)$ give an element $s \in S(I;U)$ and a certificate $C(s)$. Then the leaf elements form a finite family $s_1, \ldots, s_n$, with all the certificates $C(s_i)$ retained, and there are coefficients $a_i \in R$ such that $\sum_i s_i a_i = 1$.*

Proof. Let J be the ideal generated by the finitely many leaf elements. Every leaf supplies an element of its denominator monoid lying in J. Apply Lemma 1 upwards through the tree. At the root, $S(\varnothing;\varnothing) = \{1\}$, so $1 \in J$. The membership witness supplies the coefficients. Flattening the same finite tree retains its leaf certificates. □

There is no appeal to an infinite tree or to König's lemma. Tree formation, substitution of a finite tree at a leaf, and the proof of the theorem are finite structural inductions. If witnesses have been provided only under propositional truncation, the conclusion is also taken under truncation.

3 A decision procedure actually used by the local proof

Consider a monic list of coefficients

$$c_0, c_1, \ldots, c_{d-1}, 1.$$

Test c_0 by the two successors above. On the unit branch, stop. On the radical branch, test c_1, and continue in this way. After at most d tests the coefficient 1 terminates the process. Each leaf therefore supplies a first unit coefficient, with all earlier coefficients in the radical ideal of that leaf. This constructs the tree rather than assuming its existence.

Here is exactly why that information is enough. Let $C \to B$ be a ring map, let $\mathfrak{m} \subseteq C$ and $J \subseteq B$ be ideals with $\mathfrak{m}B \subseteq J$, and let $s \in B$ satisfy a polynomial relation modulo J. Suppose a finite prefix certificate says that the earlier coefficients lie in $\mathfrak{m}$ and the next is a unit. For a partial Horner expression, the two algebraic steps are

$$s^j(c + sH) \in J, \quad c \in \mathfrak{m} \quad \Longrightarrow \quad s^{j+1}H \in J,$$

and, when c is a unit,

$$s^j(c + sH) \in J \quad \Longrightarrow \quad s^j(1 - sv) \in J, \qquad v = -c^{-1}H.$$

Images of coefficients in B are implicit in these formulas. Reading the prefix gives $s^k(1 - sv) \in J$. Set $N = k + 1$ and $e = (sv)^N$. Then

$$e^2 - e \in J, \qquad s^N(1 - e) \in J, \qquad e = s\big(s^{N-1}v^N\big).$$

Moreover, if $s^K w \in J$, then $ew \in J$. To see the last assertion, work modulo J: e is idempotent and divisible by s, hence $e = e^K$ is divisible by s^K when $K > 0$; the case $K = 0$ is immediate. These are the relation, idempotent and annihilation properties needed by the local argument. Once the prefix certificate is present, their proof uses no further decisions.

4 Actual standard charts

Use the standard presentation

$$E \simeq \big(A[X]/(Q)\big)_{GQ'}.$$

This is expressed intrinsically by saying that maps $E \to B$ correspond to elements $x \in B$ satisfying $Q(x) = 0$ and with $(GQ')(x)$ invertible.

Proposition 3. *If E is standard for (Q, G) and x is its distinguished generator, then $E[1/F(x)]$ is standard for (Q, GF). The polynomial Q is unchanged, so monicity is retained when it is given.*

Proof. A map from this localisation to B is a map $E \to B$ for which $F(x)$ becomes a unit. The standard universal property identifies this with an element satisfying $Q(x) = 0$ and both $(GQ')(x)$ and $F(x)$ invertible. Their product is $((GF)Q')(x)$. In a commutative ring a product is a unit precisely when both factors are units. These correspondences give the required universal property. □

For a concrete two-branch example, branch on $a = F(x) \in E$. On the radical branch choose the denominator $1 - a = 1 + (-a)$; on the unit branch choose a. Proposition 3 constructs the two charts, for $(Q, G(1 - F))$ and (Q, GF). Theorem 2 supplies the certified finite cover. Neither a nor $1 - a$ needs to be a unit in E.

5 What the constructive bridge still requires

For the full local structure theorem one must construct a tree whose leaves contain chart certificates over the *original base*. The first-unit scan above is one real piece of this construction. It does not replay the whole field argument, factor selection, ZMT application, or localisation descent automatically.

In particular, a chart over a node ring must descend, with its finitely many equations and inverse-map certificates, to a suitable denominator in that node's $S(I; U)$. Descent from a localised base to a monic presentation over the original base is also a separate step. Assuming that a chart exists somewhere in the node localisation is not a substitute for giving these finite certificates. This is the precise distinction between the checked finite-tree bridge and the still unfinished global structure theorem from its original hypotheses.

Part II
Connection with the Unit development

6 Checked statements

All new code is in `unit/sag/etale-dynamical-global/`. It imports the existing constructive local and Horrocks libraries by relative links. No classical segment is imported.

- `LargeTreeCover.seg`: finite trees with leaf data in the next universe, including maps, substitution, and continuation from the existing small tree procedures. This permits actual algebras and universal properties at leaves.

- `DynamicalCertifiedCover.seg`: Theorem 2, with an explicit finite family, all certificates, and the equation $\sum_i s_i a_i = 1$, under truncation. Only the small denominator data is sent to the existing `FormalMonoidCover`; the larger certificates are retained by structural recursion. No resizing principle is needed.

- `DynamicalEtaleCover.seg`: exact chart and localisation interfaces matching the local theorem, together with a stronger monic interface. Its main theorem still takes a finite tree of such leaves.

- `FirstUnitData.seg`, `FirstUnitRelation.seg`, and `DynamicalCoefficientScan.seg`: the constructed monic scan and its interpretation in any algebra over the leaf ring, with an ideal containing the leaf radical ideal. The decision-free algebraic proofs are extracted from `etale-local/IdempotentModM.seg`; that shared file is unchanged.

- `DynamicalScanCover.seg`: compose that actual scan with a continuation constructing certified leaves. The scan is proved; the continuation is an explicit parameter.

- `StandardChartLocalisation.seg`: Proposition 3 with the full algebra universal property. A presentation and a representative for $F(x)$ construct the localisation using the existing presentation library.

- `DynamicalComplementaryCover.seg` and `DynamicalStandardCover.seg`: the two constructed branches and their actual monic standard charts. Polynomial representatives are chosen only inside the final truncation.

7 Validation and remaining integration

The fresh development checks 5,789 declarations, including 367 new declarations in eleven proof modules, in about two seconds. The `.opens` inventory covers all 367 new named declarations. Regression instances include large certificates, the two leaf positions, constant, linear and quadratic monic scans, a skipped zero coefficient, the degree-zero case over the zero ring, and actual constructed standard algebras. Deliberately changing the cover equation, the monic leading coefficient, the branch direction, or the standard denominator is rejected by the checker.

The existing univalence, replacement and surjection parameters are unchanged. There are no new axioms, no checker changes, and no edits to shared formal sources. Neither a general transformation of arbitrary local proofs nor the full global étale theorem is asserted here. The next mathematical obligation is to organise the remaining local steps into compatible finite branch procedures and descend the final chart certificates; the tree-to-cover part is now available.